The 2^{k-p} Fractional Factorial Design

- **Motivation** for fractional factorials is obvious; as the number of factors becomes large enough to be "interesting", the size of the designs grows very quickly
- Popular when the experiment involves **many** factors (often because we don't know much about the system)
- Emphasis is **screening significant effects**
- Almost always run without replicates

A Toy Example

- Three factors, each at two levels, but the experimenters cannot afford to run $2^3 = 8$ runs. They can only afford four runs.
- **Which four runs do we choose?**

TABLE 8.1

Plus and Minus Signs for the 2^3 Factorial Design

	Factorial Effect							
Treatment Combination	***I***	***A***	***B***	***C***	***AB***	***AC***	***BC***	***ABC***
a	+	+	−	−	−	−	+	+
b	+	−	+	−	−	+	−	+
c	+	−	−	+	+	−	−	+
abc	+	+	+	+	+	+	+	+
ab	+	+	+	−	+	−	−	−
ac	+	+	−	+	−	+	−	−
bc	+	−	+	+	−	−	+	−
(1)	+	−	−	−	+	+	+	−

Why is Fractional Factorial Design Possible?

- Three principles of effects
  - **Sparsity:** only a few effects are important
  - **Hierarchy:** lower-order effects are more likely to be important than higher-order effects
  - **Heredity:** an interaction can be important only if all its parent effects are also important (strong)
- We need the main effects to be estimable
- We only consider balanced designs, that is, each factor has the same number of + and –
- Which four runs do we choose?

The One-Half Fraction of the 2^3

- If we use the fraction with $I = ABC$
- The contrast for estimating the main effect A is exactly the same as the contrast used for estimating the BC interaction.
- This phenomena is called **aliasing** and it occurs in all fractional factorial designs
- Aliases can be found from the contrast table

Effects

For the example, we use the fraction with $I = ABC$

$$
\begin{aligned}
[A] &= \tfrac{1}{2}(a - b - c + abc) \\
[B] &= \tfrac{1}{2}(-a + b - c + abc) \\
[C] &= \tfrac{1}{2}(-a - b + c + abc)
\end{aligned}
$$

Aliased (fully confounded) with

$$
\begin{aligned}
[BC] &= \tfrac{1}{2}(a - b - c + abc) \\
[AC] &= \tfrac{1}{2}(-a + b - c + abc) \\
[AB] &= \tfrac{1}{2}(-a - b + c + abc)
\end{aligned}
$$

Does not mean A and BC are equal. They are not distinguishable!

Aliasing in the 2^{3-1} fractional factorial design

- **Defining relation:** $I = ABC$, ABC is called a generator or a **word**
- $I = ABC$ defines the principal fraction,

  $I = -ABC$ defines the alternate fraction
- Aliases obtained from multiplication on a word:

  For the principal fraction:

$$AI = A(ABC) = A^2BC = BC$$

$$A = BC,\ B = AC,\ C = AB$$

- We are really estimating $[I] \rightarrow I + ABC$

$$[A] \rightarrow A + BC,\quad [B] \rightarrow B + AC,\quad [C] \rightarrow C + AB$$

We are really estimating

$$[I\,] \to ABC,\ \ [A] \to A + BC,\ \ [B] \to B + AC,\ \ [C] \to C + AB$$

- I and ABC are fully confounded/ aliased;
  A and BC are aliased;
  B and AC are aliased;
  C and AB are aliased.

We cannot distinguish any pair of them.

- In this case, typically, we attribute the effect to the lower order contrast.
- We attribute $[I\,]$ to I, $[A]$ to A, $[B]$ to B, and $[C]$ to C

The Alternate Fraction of the 2^{3-1}

- $I = -ABC$ is the defining relation
- Aliasing:

$$A = -BC,\ B = -AC,\ \text{and}\ C = -AB$$

- Same aliasing as the principal fraction,
- So, the two fractions defined by

$$I = \pm ABC$$

have the same statistical property

Design Resolution

- Resolution: length of the word, that is, the number of factors involved in the word. When a design has multiple words, resolution is the length of the shortest word
- A design is of resolution R if no p-factor effect is aliased with another effect containing less than $R-p$ factors.

- Resolution III Designs:
  $me = 2fi$
  example 2_{III}^{3-1}

- Resolution IV Designs:
  $2fi = 2fi$
  example 2_{IV}^{4-1}

- Resolution V Designs:
  $2fi = 3fi$
  example 2_{V}^{5-1}

2_{IV}^{4-1} Design

Run	I	A	B	C	D	AB	AC	AD	BC	BD	CD	ABC	ABD	ACD	BCD	ABCD
1	+	−	−	−	−	+	+	+	+	+	+	−	−	−	−	+
2	+	−	−	+	+	+	−	−	−	−	+	+	+	−	−	+
3	+	−	+	−	+	−	+	−	−	+	−	+	−	+	−	+
4	+	−	+	+	−	−	−	+	+	−	−	−	+	+	−	+
5	+	+	−	−	+	−	−	+	+	−	−	+	−	−	+	+
6	+	+	−	+	−	−	+	−	−	+	−	−	+	−	+	+
7	+	+	+	−	−	+	−	−	−	−	+	−	−	+	+	+
8	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+
9	+	−	−	−	+	+	+	−	+	−	−	−	+	+	+	−
10	+	−	−	+	−	+	−	+	−	+	−	+	−	+	+	−
11	+	−	+	−	−	−	+	+	−	−	+	+	+	−	+	−
12	+	−	+	+	+	−	−	−	+	+	+	−	−	−	+	−
13	+	+	−	−	−	−	−	−	+	+	+	+	+	+	−	−
14	+	+	−	+	+	−	+	+	−	−	+	−	−	+	−	−
15	+	+	+	−	+	+	−	+	−	+	−	−	+	−	−	−
16	+	+	+	+	−	+	+	−	+	−	−	+	−	−	−	−

Construction of a One-half Fraction

The **basic** design; the design **generator**

■ **TABLE 8.2**

The Two One-Half Fractions of the 2^3 Design

	Full 2^2 Factorial (Basic Design)		2_{III}^{3-1}, $I = ABC$			2_{III}^{3-1}, $I = -ABC$		
Run	A	B	A	B	$C = AB$	A	B	$C = -AB$
1	−	−	−	−	+	−	−	−
2	+	−	+	−	−	+	−	+
3	−	+	−	+	−	−	+	+
4	+	+	+	+	+	+	+	−